

Week 5: GANs Kaggle: I'm Something of a Painter Myself

Project Topic: Train GANs to generate 7000 - 10000 Monet-Style images.

1. Import the necessary python libraries

```
In [1]: import os
import zipfile
import glob
import time
import numpy as np
from PIL import Image
import tensorflow as tf
from tensorflow.keras import layers
import matplotlib.pyplot as plt
```

```
2025-04-24 03:49:45.110693: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2025-04-24 03:49:45.139628: E external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:485] Unable to register cuFFT factory: Attempting to register factory for plugin cuFFT when one has already been registered
2025-04-24 03:49:45.153530: E external/local_xla/xla/stream_executor/cuda/cuda_dnn.cc:8473] Unable to register cuDNN factory: Attempting to register factory for plugin cuDNN when one has already been registered
2025-04-24 03:49:45.157091: E external/local_xla/xla/stream_executor/cuda/cuda_blas.cc:1471] Unable to register cuBLAS factory: Attempting to register factory for plugin cuBLAS when one has already been registered
2025-04-24 03:49:45.177171: I tensorflow/core/platform/cpu_feature_guard.cc:211] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX, in other operations, rebuild TensorFlow with the appropriate compiler flags.
```

This snippet tells TensorFlow to only grab GPU memory as it's needed, instead of reserving all of it up front

```
In [2]: for gpu in tf.config.list_physical_devices('GPU'):
        tf.config.experimental.set_memory_growth(gpu, True)
```

```
WARNING: All log messages before absl::InitializeLog() is called are written
to STDERR
I0000 00:00:1745466587.929783    9150 cuda_executor.cc:1001] could not open
file to read NUMA node: /sys/bus/pci/devices/0000:01:00.0/numa_node
Your kernel may have been built without NUMA support.
I0000 00:00:1745466588.392091    9150 cuda_executor.cc:1001] could not open
file to read NUMA node: /sys/bus/pci/devices/0000:01:00.0/numa_node
Your kernel may have been built without NUMA support.
I0000 00:00:1745466588.392144    9150 cuda_executor.cc:1001] could not open
file to read NUMA node: /sys/bus/pci/devices/0000:01:00.0/numa_node
Your kernel may have been built without NUMA support.
```

Get the paths for JPGs and TIFs files.

```
In [3]: monet_data_dir = os.path.join(os.getcwd(), "gan-getting-started")
monet_jpg_dir = os.path.join(monet_data_dir, "monet_jpg")
monet_tfrec_dir = os.path.join(monet_data_dir, "monet_tfrec")
photo_jpg_dir = os.path.join(monet_data_dir, "photo_jpg")
photo_tfrec_dir = os.path.join(monet_data_dir, "photo_tfrec")
```

2 Exploratory Data Analysis (EDA)

Prints the total number of Monet images, and the first 5 paths of the Monet images.

```
In [4]: monet_jpgs = [os.path.join(monet_jpg_dir, jpg) for jpg in os.listdir(monet_jpg_dir)]

print(f"Total number of Monet images: {len(monet_jpgs)}")
print(f"First 5 images:\n{monet_jpgs[:5]}")
```

Total number of Monet images: 300

First 5 images:

```
['/workspace/gan-getting-started/monet_jpg/632ddbc784.jpg', '/workspace/gan-getting-started/monet_jpg/a8fbb3eb1.jpg', '/workspace/gan-getting-started/monet_jpg/fdf1530d95.jpg', '/workspace/gan-getting-started/monet_jpg/7239ba0b55.jpg', '/workspace/gan-getting-started/monet_jpg/7341d96c1d.jpg']
```

Prints the total number of Photo images, and the first 5 paths of the Photo images.

```
In [5]: photo_jpgs = [os.path.join(photo_jpg_dir, jpg) for jpg in os.listdir(photo_jpg_dir)]

print(f"Total number of Photo images: {len(photo_jpgs)}")
print(f"First 5 images:\n{photo_jpgs[:5]}")
```

Total number of Photo images: 7038

First 5 images:

```
['/workspace/gan-getting-started/photo_jpg/472d5ac100.jpg', '/workspace/gan-getting-started/photo_jpg/716981b92d.jpg', '/workspace/gan-getting-started/photo_jpg/d3f0f026a8.jpg', '/workspace/gan-getting-started/photo_jpg/c0b977acfb.jpg', '/workspace/gan-getting-started/photo_jpg/41f062b74b.jpg']
```

Sets Global variables to be used later on.

- `IMG_SIZE = 256`: Resize all input images to 256×256 pixels.
- `BATCH_SIZE = 4`: Process four images at a time during training.
- `EPOCHS = 100`: Run 100 full passes over the dataset.
- `LR = 2e-4`: Use a learning rate of 0.0002 for both generators and discriminators.
- `BETA_1 = 0.5`: Set the Adam optimizer's exponential decay rate for the first moment estimates.
- `LAMBDA_CYCLE = 10.0`: Weight of the cycle-consistency loss term.
- `LAMBDA_ID = 0.5 * LAMBDA_CYCLE`: Weight of the identity loss (half the cycle weight).
- `SAMPLE_INTERVAL = 10`: Save example translations every 10 epochs.
- `OUTPUT_DIR = "cyclegan_output"`: Folder for generated images, model checkpoints, etc.; `os.makedirs(..., exist_ok=True)` creates it if it doesn't already exist.
- `AUTOTUNE = tf.data.AUTOTUNE`: Let TensorFlow dynamically tune parallel calls and prefetch buffer sizes for your data pipeline.

```
In [6]: IMG_SIZE      = 256
        BATCH_SIZE   = 4
        EPOCHS       = 100
        LR           = 2e-4
        BETA_1        = 0.5
        LAMBDA_CYCLE  = 10.0
        LAMBDA_ID     = 0.5 * LAMBDA_CYCLE
        SAMPLE_INTERVAL = 10
        OUTPUT_DIR    = "cyclegan_output"
        AUTOTUNE      = tf.data.AUTOTUNE
        os.makedirs(OUTPUT_DIR, exist_ok=True)
```

This function reads a JPEG file from disk, decodes it into a 3-channel image tensor, and resizes it to your model's fixed `IMG_SIZE` square. Finally, it scales the pixel values from the original `[0, 255]` range to `[-1, 1]`, which helps stabilize GAN training, and returns the ready-to-use image tensor.

```
In [7]: def load_image(path):
        img = tf.io.read_file(path)
        img = tf.image.decode_jpeg(img, channels=3)
        img = tf.image.resize(img, [IMG_SIZE, IMG_SIZE])
        img = (img / 127.5) - 1.0
        return img
```

Prints out the shape of the a sample Monet JPG.

```
In [8]: sample_monet_jpg = load_image(monet_jpgs[0])
        print(f"JPG shape: {sample_monet_jpg.shape}")
```

JPG shape: (256, 256, 3)

```
I0000 00:00:1745466588.648247    9150 cuda_executor.cc:1001] could not open
file to read NUMA node: /sys/bus/pci/devices/0000:01:00.0/numa_node
Your kernel may have been built without NUMA support.
I0000 00:00:1745466588.648302    9150 cuda_executor.cc:1001] could not open
file to read NUMA node: /sys/bus/pci/devices/0000:01:00.0/numa_node
Your kernel may have been built without NUMA support.
I0000 00:00:1745466588.648312    9150 cuda_executor.cc:1001] could not open
file to read NUMA node: /sys/bus/pci/devices/0000:01:00.0/numa_node
Your kernel may have been built without NUMA support.
I0000 00:00:1745466588.806609    9150 cuda_executor.cc:1001] could not open
file to read NUMA node: /sys/bus/pci/devices/0000:01:00.0/numa_node
Your kernel may have been built without NUMA support.
I0000 00:00:1745466588.806657    9150 cuda_executor.cc:1001] could not open
file to read NUMA node: /sys/bus/pci/devices/0000:01:00.0/numa_node
Your kernel may have been built without NUMA support.
2025-04-24 03:49:48.806667: I tensorflow/core/common_runtime/gpu/gpu_device.
cc:2112] Could not identify NUMA node of platform GPU id 0, defaulting to 0.
Your kernel may not have been built with NUMA support.
I0000 00:00:1745466588.806698    9150 cuda_executor.cc:1001] could not open
file to read NUMA node: /sys/bus/pci/devices/0000:01:00.0/numa_node
Your kernel may have been built without NUMA support.
2025-04-24 03:49:48.806739: I tensorflow/core/common_runtime/gpu/gpu_device.
cc:2021] Created device /job:localhost/replica:0/task:0/device:GPU:0 with 26
747 MB memory:  -> device: 0, name: NVIDIA GeForce RTX 5090, pci bus id: 000
0:01:00.0, compute capability: 12.0
```

This code creates a 3×3 grid of 15×15-inch subplots at 100 DPI, then flattens the axes array so it can loop through the first nine Monet image paths. For each subplot, it loads and rescales the image tensor from [-1, 1] back to [0, 1] for display, hides the axis ticks, and titles it with its image number. Finally, it applies a tight layout to remove extra padding and renders the figure.

```
In [9]: fig, axes = plt.subplots(
        nrows=3, ncols=3,
        figsize=(15, 15),    # square canvas
        dpi=100
    )

    axes = axes.flatten()
    for ax, path in zip(axes, monet_jpgs[:9]):
        img = load_image(path)
        ax.imshow(((img + 1.0) / 2.0).numpy())
        ax.axis("off")
        ax.set_title(f"Image {monet_jpgs.index(path) + 1}")

    plt.tight_layout()
    plt.show()
```




Prints out the shape of the a sample Photo JPG.

```
In [10]: sample_photo_jpg = load_image(photo_jpgs[0])
print(f"JPG shape: {sample_photo_jpg.shape}")
```

JPG shape: (256, 256, 3)

This code creates a 3×3 grid of 15×15-inch subplots at 100 DPI, then flattens the axes array so it can loop through the first nine Photo image paths. For each subplot, it loads and rescales the image tensor from $[-1, 1]$ back to $[0, 1]$ for display, hides the axis ticks, and titles it with its image number. Finally, it applies a tight layout to remove extra padding and renders the figure.

```
In [11]: fig, axes = plt.subplots(
    nrows=3, ncols=3,
    figsize=(15, 15),    # square canvas
    dpi=100)
```

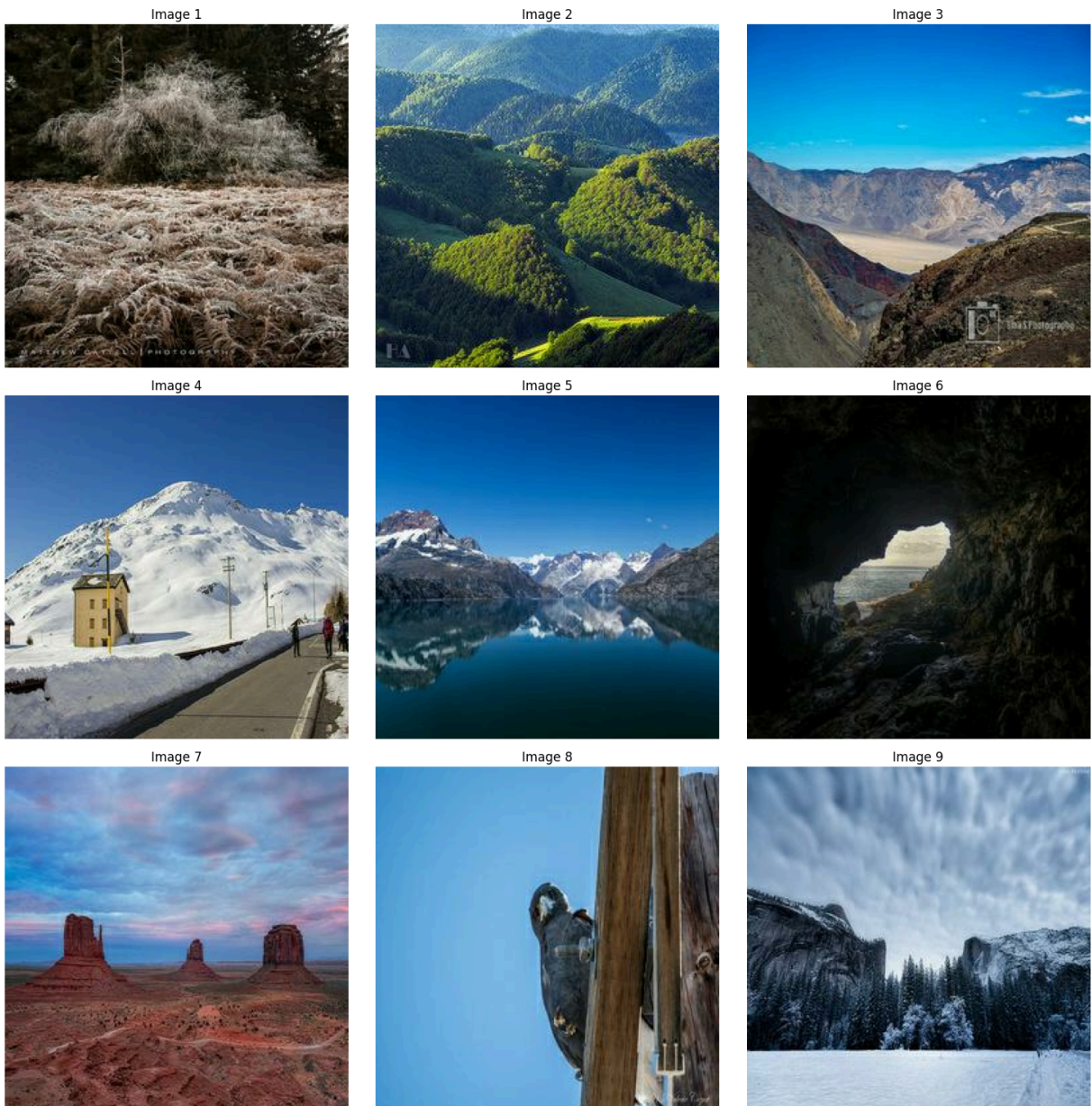
```

)

axes = axes.flatten()
for ax, path in zip(axes, photo_jpgs[:9]):
    img = load_image(path)
    ax.imshow(((img + 1.0) / 2.0).numpy())
    ax.axis("off")
    ax.set_title(f"Image {photo_jpgs.index(path) + 1}")

plt.tight_layout()
plt.show()

```



Model Buliding & Architecture

This defines a helper that turns a list of file paths into a fully-pipeline'd `tf.data.Dataset`: it slices the paths, maps `load_image` across them in parallel, shuffles the examples, groups them into batches of size `BATCH_SIZE`, and

prefetches with AUTOTUNE for performance. Finally, it instantiates two datasets—`monet_ds` and `photo_ds`—from your Monet and photo image path lists.

```
In [12]: def make_dataset(paths):
          ds = tf.data.Dataset.from_tensor_slices(paths)
          ds = ds.map(load_image, num_parallel_calls=AUTOTUNE)
          ds = ds.shuffle(len(paths)).batch(BATCH_SIZE).prefetch(AUTOTUNE)
          return ds

          monet_ds = make_dataset(monet_jpgs)
          photo_ds = make_dataset(photo_jpgs)
```

This function implements a single ResNet “skip” block: it takes an input tensor `x`, applies two 3×3 convolutions (without bias) each followed by batch normalization and a ReLU (after the first convolution), then adds the original tensor (`init`) back to the result, enabling the network to learn residual mappings and improving gradient flow during training.

```
In [13]: def resnet_block(x, filters):
          init = x
          x = layers.Conv2D(filters, 3, padding='same', use_bias=False)(x)
          x = layers.BatchNormalization()(x)
          x = layers.ReLU()(x)

          x = layers.Conv2D(filters, 3, padding='same', use_bias=False)(x)
          x = layers.BatchNormalization()(x)
          return layers.add([init, x])
```

This function builds the CycleGAN generator as a Keras Model that:

- Accepts a $256 \times 256 \times 3$ input and applies a 7×7 convolution (64 filters) with batch normalization and ReLU.
- Downsamples twice via 3×3 convolutions (128 then 256 filters) with stride 2, each followed by BN+ReLU.
- Processes features through nine ResNet blocks (each performing two 3×3 convs + BN + ReLU and a skip connection) at 256 channels.
- Upsamples back to 256×256 with two 3×3 transpose convolutions (128 then 64 filters), again with BN+ReLU.
- Outputs a $256 \times 256 \times 3$ image via a final 7×7 convolution with tanh, producing pixel values in $[-1, 1]$.

```
In [14]: def build_generator():
          inputs = layers.Input(shape=(IMG_SIZE, IMG_SIZE, 3))
          x = layers.Conv2D(64, 7, padding='same', use_bias=False)(inputs)
          x = layers.BatchNormalization()(x)
          x = layers.ReLU()(x)
```

```

for f in [128, 256]:
    x = layers.Conv2D(f, 3, strides=2, padding='same', use_bias=False)(x)
    x = layers.BatchNormalization()(x)
    x = layers.ReLU()(x)

for _ in range(9):
    x = resnet_block(x, 256)

for f in [128, 64]:
    x = layers.Conv2DTranspose(f, 3, strides=2, padding='same', use_bias=False)(x)
    x = layers.BatchNormalization()(x)
    x = layers.ReLU()(x)

x = layers.Conv2D(3, 7, padding='same', activation='tanh')(x)
return tf.keras.Model(inputs, x, name="Generator")

```

Call the `build_generator()` function to build a generator for Photo2Monet and Monet2Photo

```

In [15]: G_photo2monet = build_generator()
         G_monet2photo = build_generator()

```

This defines the CycleGAN discriminator as a Keras Model that takes a 256×256×3 image input and applies a series of 4×4 convolutions with increasing filters (64→128→256→512), downsampling via strides of 2 (except the last block), each followed by a LeakyReLU activation ($\alpha=0.2$). A final 4×4 convolution projects to a single-channel output map, yielding a “PatchGAN” score for each patch to distinguish real from generated patches.

```

In [16]: def build_discriminator():
         inp = layers.Input(shape=(IMG_SIZE, IMG_SIZE, 3))
         x = inp
         for f, stride in zip([64, 128, 256, 512], [2, 2, 2, 1]):
             x = layers.Conv2D(f, 4, strides=stride, padding='same')(x)
             x = layers.LeakyReLU(0.2)(x)
         x = layers.Conv2D(1, 4, padding='same')(x)
         return tf.keras.Model(inp, x, name="Discriminator")

```

Call the `build_discriminator()` function to build a discriminator for Monet and Photo images.

```

In [17]: D_monet = build_discriminator()
         D_photo = build_discriminator()

```

This code cell sets up the loss functions and optimizers in one go. Use `MeanSquaredError` for adversarial terms and `MeanAbsoluteError` for cycle-consistency and identity losses. The `adversarial_loss` pushes generated images’ predictions toward 1, while `discriminator_loss` averages the penalty for real (1) and fake (0) inputs. The `cycle_loss` measures absolute differences between

originals and cycled images (scaled by LAMBDA_CYCLE), and identity_loss does the same for identity mappings (scaled by LAMBDA_ID). Finally, create Adam optimizers for both generator and discriminator using the learning rate (LR) and momentum term (BETA_1).

```
In [18]: adv_loss_fn = tf.keras.losses.MeanSquaredError()
cycle_loss_fn = tf.keras.losses.MeanAbsoluteError()

def adversarial_loss(real, pred):
    return adv_loss_fn(tf.ones_like(pred), pred)

def discriminator_loss(real, fake):
    real_loss = adv_loss_fn(tf.ones_like(real), real)
    fake_loss = adv_loss_fn(tf.zeros_like(fake), fake)
    return 0.5 * (real_loss + fake_loss)

def cycle_loss(real, cycled):
    return cycle_loss_fn(real, cycled) * LAMBDA_CYCLE

def identity_loss(real, same):
    return cycle_loss_fn(real, same) * LAMBDA_ID

gen_opt = tf.keras.optimizers.Adam(LR, BETA_1)
disc_opt = tf.keras.optimizers.Adam(LR, BETA_1)
```

This snippet first creates two separate Adam optimizers—disc_monet_opt and disc_photo_opt—both using shared learning rate and momentum term. Then it defines a @tf.function-decorated train_step that, in one go, generates fake Monet and photo images, cycles them back, and computes the identity mappings; it evaluates all real and fake images with the respective discriminators, assembles the adversarial, cycle-consistency, and identity losses into a total generator loss, and computes discriminator losses. Finally, it takes gradients of those losses with respect to each network’s weights, applies the corresponding optimizer updates, and returns a dictionary of the generator and both discriminator losses for logging.

```
In [19]: disc_monet_opt = tf.keras.optimizers.Adam(LR, beta_1=BETA_1)
disc_photo_opt = tf.keras.optimizers.Adam(LR, beta_1=BETA_1)

@tf.function
def train_step(real_monet, real_photo):
    with tf.GradientTape(persistent=True) as tape:
        fake_monet = G_photo2monet(real_photo, training=True)
        fake_photo = G_monet2photo(real_monet, training=True)

        cycled_photo = G_monet2photo(fake_monet, training=True)
        cycled_monet = G_photo2monet(fake_photo, training=True)

        same_photo = G_monet2photo(real_photo, training=True)
        same_monet = G_photo2monet(real_monet, training=True)
```

```

disc_real_monet = D_monet(real_monet, training=True)
disc_real_photo = D_photo(real_photo, training=True)
disc_fake_monet = D_monet(fake_monet, training=True)
disc_fake_photo = D_photo(fake_photo, training=True)

gen_g_adv = adversarial_loss(disc_fake_monet, disc_fake_monet)
gen_f_adv = adversarial_loss(disc_fake_photo, disc_fake_photo)
total_cycle = cycle_loss(real_photo, cycled_photo) + cycle_loss(real_monet, cycled_monet)
id_loss = identity_loss(real_photo, same_photo) + identity_loss(real_monet, same_monet)
G_loss = gen_g_adv + gen_f_adv + total_cycle + id_loss

D_monet_loss = discriminator_loss(disc_real_monet, disc_fake_monet)
D_photo_loss = discriminator_loss(disc_real_photo, disc_fake_photo)

grads_G = tape.gradient(G_loss, G_photo2monet.trainable_variables + G_monet.trainable_variables)
grads_D_m = tape.gradient(D_monet_loss, D_monet.trainable_variables)
grads_D_p = tape.gradient(D_photo_loss, D_photo.trainable_variables)

gen_opt.apply_gradients(zip(grads_G, G_photo2monet.trainable_variables + G_monet.trainable_variables))
disc_monet_opt.apply_gradients(zip(grads_D_m, D_monet.trainable_variables))
disc_photo_opt.apply_gradients(zip(grads_D_p, D_photo.trainable_variables))

return {
    "G_loss": G_loss,
    "D_monet_loss": D_monet_loss,
    "D_photo_loss": D_photo_loss
}

```

This loop first pulls one batch of real photos to use as a fixed “example” input. It then runs through EPOCHS iterations, and in each epoch it zips together the Monet and photo datasets, calling train_step on every pair to update the generators and discriminators. After each epoch it prints out the formatted generator and discriminator losses along with the elapsed time. On the first epoch and every SAMPLE_INTERVALth epoch thereafter, it feeds that fixed photo batch through the photo→Monet generator (inference mode), rescales the outputs back to 0-255, arranges them in a 1×BATCH_SIZE subplot row, titles the figure with the epoch number, and saves it to the OUTPUT_DIR before closing the plot.

```

In [20]: example_photo = next(iter(photo_ds))

for epoch in range(1, EPOCHS+1):
    start = time.time()
    for real_monet, real_photo in tf.data.Dataset.zip((monet_ds, photo_ds)):
        losses = train_step(real_monet, real_photo)
    print(f"Epoch {epoch:03d}/{EPOCHS} "
          f"G_loss: {losses['G_loss']:.4f} "
          f"D_m: {losses['D_monet_loss']:.4f} "
          f"D_p: {losses['D_photo_loss']:.4f} "
          f"Time: {time.time()-start:.1f}s")

    if epoch % SAMPLE_INTERVAL == 0 or epoch == 1:
        fake = G_photo2monet(example_photo, training=False)

```

```
fake = (fake + 1) * 127.5
fake = tf.cast(fake, tf.uint8).numpy()
fig, axes = plt.subplots(1, BATCH_SIZE, figsize=(16,4))
for i, ax in enumerate(axes):
    ax.imshow(fake[i])
    ax.axis('off')
plt.suptitle(f"Photo→Monet at Epoch {epoch}")
plt.savefig(os.path.join(OUTPUT_DIR, f"sample_epoch_{epoch:03d}.png"))
plt.close()
```

```
2025-04-24 03:50:04.387298: I external/local_xla/xla/stream_executor/cuda/cuda_dnn.cc:531] Loaded cuDNN version 90701
W0000 00:00:1745466604.505251    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.523071    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.523222    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.523498    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.523642    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.524025    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.524255    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.536822    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.536979    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.537366    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.538410    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.538565    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.538746    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.538976    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.539184    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.544568    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.544743    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.546471    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.546670    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.547407    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.547587    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.547775    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.547982    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.558787    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.570525    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.570797    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.571107    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
```


[illegible]

```
W0000 00:00:1745466604.585996    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.586287    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.586632    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.587050    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.587366    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.587757    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.588153    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.588602    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.588991    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.589442    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.589917    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.590388    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.590845    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.591502    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.591994    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.593273    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.594332    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
W0000 00:00:1745466604.776786    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.777029    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.777483    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.777764    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.777960    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.778268    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.778467    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466604.778661    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
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[illegible]


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W0000 00:00:1745466606.308462    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466606.308717    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466606.308947    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1745466606.311266    9228 service.cc:146] XLA service 0x7f33c92fc2d0 initialized for platform CUDA (this does not guarantee that XLA will be used). Devices:
I0000 00:00:1745466606.311359    9228 service.cc:154] StreamExecutor device (0): NVIDIA GeForce RTX 5090, Compute Capability 12.0
2025-04-24 03:50:06.316384: I tensorflow/compiler/mlir/tensorflow/utils/dump_mlir_util.cc:268] disabling MLIR crash reproducer, set env var `MLIR_CRASH_REPRODUCER_DIRECTORY` to enable.
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
I0000 00:00:1745466606.375157    9228 device_compiler.h:188] Compiled cluster using XLA! This line is logged at most once for the lifetime of the process.
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
W0000 00:00:1745466606.422533    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466606.422907    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466606.423347    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466606.423801    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466606.424142    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466606.424525    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466606.424927    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466606.425360    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466606.425730    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466606.426100    9228 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466606.426450    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466606.426925    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
W0000 00:00:1745466606.427503    9230 gpu_timer.cc:114] Skipping the delay kernel, measurement accuracy will be reduced
```

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ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.445076 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.445585 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.445927 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.446629 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.446961 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.447629 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.448170 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.450247 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.450845 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.451840 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.452328 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.453025 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.454278 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.456343 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.456778 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.457228 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.457931 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.458559 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.459063 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.460031 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.460554 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.461061 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.461607 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.462412 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.463117 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
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'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
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'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
W0000 00:00:1745466606.566615    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.566848    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.567041    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.567251    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.567464    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.567811    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.568364    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.568693    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.568910    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.569157    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.569389    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.569633    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.569896    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.570151    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.570397    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.570690    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.570956    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.571229    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.571550    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.571837    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.572119    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.572374    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.572648    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.572923    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.573204    9228 gpu_timer.cc:114] Skipping the delay k
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ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.573547 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.573876 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.574207 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.574532 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.574866 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.575222 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.577155 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
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'+ptx85' is not a recognized feature for this target (ignoring feature)
W0000 00:00:1745466606.667726 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.667932 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.668208 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.668579 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.668945 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.669278 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.669693 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.669951 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.670194 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.670447 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.670706 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.671070 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.671339 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.671612 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.671885 9228 gpu_timer.cc:114] Skipping the delay k
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ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.700712 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.702380 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.702724 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.703641 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.703970 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.705006 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.705518 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.706865 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.709279 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
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'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
W0000 00:00:1745466606.771824 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.772121 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.773039 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.773508 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.773665 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.773901 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.774073 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.774365 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.774743 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.775087 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.775442 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.775773 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.776068 9228 gpu_timer.cc:114] Skipping the delay k
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ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.776361 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.776719 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.777103 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.778187 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.778505 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.780318 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.780618 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.781497 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.781842 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.785250 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.787059 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.789335 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.790365 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
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'+ptx85' is not a recognized feature for this target (ignoring feature)
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'+ptx85' is not a recognized feature for this target (ignoring feature)
W0000 00:00:1745466606.878475 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.879434 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.879760 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.880059 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.880375 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.880667 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.881014 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.881336 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.881740 9230 gpu_timer.cc:114] Skipping the delay k
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ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.900194 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.900804 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.903635 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.905885 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.908466 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.911299 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.919739 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.930203 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.930462 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.930745 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.931063 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.931356 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.931637 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.931987 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.932284 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.932654 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.933024 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.933384 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.933759 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.934092 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.934440 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.934872 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.935205 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.935615 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.936022 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.936829 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.938848 9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
'+ptx85' is not a recognized feature for this target (ignoring feature)
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'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
'+ptx85' is not a recognized feature for this target (ignoring feature)
W0000 00:00:1745466606.988314    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.989839    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.991033    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.992372    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.993693    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.995228    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.996748    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.998337    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466606.999942    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466607.001501    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466607.003092    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466607.004951    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466607.006235    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466607.009490    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466607.012737    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466607.015249    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466607.017738    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466607.024243    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466607.031535    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466607.032599    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466607.032920    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466607.033369    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466607.033814    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466607.034292    9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466607.034746    9228 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466607.035243    9230 gpu_timer.cc:114] Skipping the delay k
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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

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ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.390230 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.390527 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.390800 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.391131 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.392181 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.394012 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.394760 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.399298 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.399597 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.399727 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.399933 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.400133 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.400407 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.400663 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.400957 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.401221 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.401537 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.402587 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.404583 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466609.405307 9230 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
2025-04-24 03:50:21.251762: I tensorflow/core/framework/local_rendezvous.cc:
404] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence
W0000 00:00:1745466621.298026 9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466621.298285 9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466621.298488 9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466621.305548 9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466621.306053 9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466621.306606 9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466621.307303 9150 gpu_timer.cc:114] Skipping the delay k
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[illegible]

[illegible]

[illegible]

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ernel, measurement accuracy will be reduced
W0000 00:00:1745466621.420501    9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466621.423112    9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745466621.425779    9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
Epoch 001/100  G_loss: 6.0448  D_m: 0.2998  D_p: 0.3196  Time: 27.6s
2025-04-24 03:50:33.951583: I tensorflow/core/framework/local_rendezvous.cc:
404] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence
Epoch 002/100  G_loss: 5.8431  D_m: 0.2030  D_p: 0.2605  Time: 12.4s
Epoch 003/100  G_loss: 5.8504  D_m: 0.2093  D_p: 0.2379  Time: 12.6s
2025-04-24 03:50:58.900336: I tensorflow/core/framework/local_rendezvous.cc:
404] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence
Epoch 004/100  G_loss: 5.8763  D_m: 0.2013  D_p: 0.2127  Time: 12.4s
Epoch 005/100  G_loss: 5.6920  D_m: 0.1953  D_p: 0.2013  Time: 12.5s
Epoch 006/100  G_loss: 7.7275  D_m: 0.2287  D_p: 0.2393  Time: 12.4s
Epoch 007/100  G_loss: 7.2662  D_m: 0.2477  D_p: 0.2125  Time: 12.5s
2025-04-24 03:51:48.919567: I tensorflow/core/framework/local_rendezvous.cc:
404] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence
Epoch 008/100  G_loss: 6.1344  D_m: 0.1886  D_p: 0.2263  Time: 12.6s
Epoch 009/100  G_loss: 4.7953  D_m: 0.1681  D_p: 0.1960  Time: 12.4s
Epoch 010/100  G_loss: 5.8340  D_m: 0.2156  D_p: 0.1781  Time: 12.5s
Epoch 011/100  G_loss: 5.4893  D_m: 0.1783  D_p: 0.1797  Time: 12.5s
Epoch 012/100  G_loss: 5.2920  D_m: 0.1285  D_p: 0.2634  Time: 12.4s
Epoch 013/100  G_loss: 5.8380  D_m: 0.1521  D_p: 0.1677  Time: 12.4s
Epoch 014/100  G_loss: 6.8224  D_m: 0.1483  D_p: 0.1742  Time: 12.3s
Epoch 015/100  G_loss: 8.2518  D_m: 0.1537  D_p: 0.1603  Time: 12.5s
2025-04-24 03:53:28.387480: I tensorflow/core/framework/local_rendezvous.cc:
404] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence
Epoch 016/100  G_loss: 4.8850  D_m: 0.1651  D_p: 0.2191  Time: 12.3s
Epoch 017/100  G_loss: 4.8921  D_m: 0.1376  D_p: 0.1487  Time: 12.5s
Epoch 018/100  G_loss: 5.0613  D_m: 0.1462  D_p: 0.2088  Time: 12.3s
Epoch 019/100  G_loss: 5.4913  D_m: 0.1480  D_p: 0.2308  Time: 12.4s
Epoch 020/100  G_loss: 5.5949  D_m: 0.1176  D_p: 0.1212  Time: 12.5s
Epoch 021/100  G_loss: 6.5777  D_m: 0.1840  D_p: 0.1753  Time: 12.3s
Epoch 022/100  G_loss: 4.6074  D_m: 0.1698  D_p: 0.1871  Time: 12.4s
Epoch 023/100  G_loss: 4.3979  D_m: 0.1305  D_p: 0.1537  Time: 12.3s
Epoch 024/100  G_loss: 5.0612  D_m: 0.1169  D_p: 0.1425  Time: 12.5s
Epoch 025/100  G_loss: 4.4402  D_m: 0.0912  D_p: 0.1594  Time: 12.4s
Epoch 026/100  G_loss: 5.1869  D_m: 0.0882  D_p: 0.1749  Time: 12.4s
Epoch 027/100  G_loss: 4.8052  D_m: 0.1075  D_p: 0.1175  Time: 12.5s
Epoch 028/100  G_loss: 4.9028  D_m: 0.0723  D_p: 0.1883  Time: 12.4s
Epoch 029/100  G_loss: 4.5312  D_m: 0.1759  D_p: 0.2087  Time: 12.6s
Epoch 030/100  G_loss: 4.6521  D_m: 0.2109  D_p: 0.1994  Time: 12.4s
Epoch 031/100  G_loss: 4.7156  D_m: 0.0989  D_p: 0.1129  Time: 12.4s
2025-04-24 03:56:47.553731: I tensorflow/core/framework/local_rendezvous.cc:
404] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence
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Epoch 032/100	G_loss: 6.4013	D_m: 0.1021	D_p: 0.1443	Time: 12.5s
Epoch 033/100	G_loss: 6.1450	D_m: 0.1952	D_p: 0.1870	Time: 12.4s
Epoch 034/100	G_loss: 4.6477	D_m: 0.2107	D_p: 0.1391	Time: 12.6s
Epoch 035/100	G_loss: 4.4034	D_m: 0.1099	D_p: 0.1128	Time: 12.4s
Epoch 036/100	G_loss: 5.1442	D_m: 0.1213	D_p: 0.1368	Time: 20.5s
Epoch 037/100	G_loss: 6.2595	D_m: 0.1031	D_p: 0.0942	Time: 12.3s
Epoch 038/100	G_loss: 6.2160	D_m: 0.1163	D_p: 0.1362	Time: 12.5s
Epoch 039/100	G_loss: 5.0497	D_m: 0.1181	D_p: 0.1248	Time: 12.4s
Epoch 040/100	G_loss: 4.5697	D_m: 0.0771	D_p: 0.1818	Time: 12.4s
Epoch 041/100	G_loss: 4.6468	D_m: 0.1082	D_p: 0.1371	Time: 12.5s
Epoch 042/100	G_loss: 3.8551	D_m: 0.0918	D_p: 0.1535	Time: 12.3s
Epoch 043/100	G_loss: 4.4958	D_m: 0.1400	D_p: 0.1186	Time: 12.5s
Epoch 044/100	G_loss: 4.7556	D_m: 0.0757	D_p: 0.1983	Time: 12.3s
Epoch 045/100	G_loss: 4.7426	D_m: 0.1365	D_p: 0.1373	Time: 12.5s
Epoch 046/100	G_loss: 4.2035	D_m: 0.0806	D_p: 0.2795	Time: 12.5s
Epoch 047/100	G_loss: 4.3882	D_m: 0.1828	D_p: 0.1300	Time: 12.4s
Epoch 048/100	G_loss: 5.1683	D_m: 0.1479	D_p: 0.1562	Time: 12.5s
Epoch 049/100	G_loss: 4.0518	D_m: 0.1862	D_p: 0.2119	Time: 12.4s
Epoch 050/100	G_loss: 4.3674	D_m: 0.1699	D_p: 0.1601	Time: 12.5s
Epoch 051/100	G_loss: 5.2297	D_m: 0.1320	D_p: 0.0974	Time: 12.4s
Epoch 052/100	G_loss: 5.1821	D_m: 0.1242	D_p: 0.2300	Time: 12.4s
Epoch 053/100	G_loss: 4.6667	D_m: 0.1139	D_p: 0.1102	Time: 12.6s
Epoch 054/100	G_loss: 4.3907	D_m: 0.1547	D_p: 0.1577	Time: 12.3s
Epoch 055/100	G_loss: 3.6989	D_m: 0.1409	D_p: 0.1870	Time: 12.5s
Epoch 056/100	G_loss: 5.6262	D_m: 0.1798	D_p: 0.1503	Time: 12.4s
Epoch 057/100	G_loss: 5.2838	D_m: 0.1485	D_p: 0.1628	Time: 12.5s
Epoch 058/100	G_loss: 4.1790	D_m: 0.1679	D_p: 0.1318	Time: 12.5s
Epoch 059/100	G_loss: 4.8257	D_m: 0.1174	D_p: 0.1640	Time: 12.4s
Epoch 060/100	G_loss: 3.7993	D_m: 0.2524	D_p: 0.2215	Time: 12.6s
Epoch 061/100	G_loss: 4.5406	D_m: 0.1449	D_p: 0.1666	Time: 12.3s
Epoch 062/100	G_loss: 4.4244	D_m: 0.1819	D_p: 0.1608	Time: 12.5s
Epoch 063/100	G_loss: 4.3935	D_m: 0.1514	D_p: 0.2076	Time: 12.5s

2025-04-24 04:03:33.775131: I tensorflow/core/framework/local_rendezvous.cc:404] Local rendezvous is aborting with status: OUT_OF_RANGE: End of sequence

Epoch 064/100	G_loss: 4.1244	D_m: 0.1281	D_p: 0.2403	Time: 12.5s
Epoch 065/100	G_loss: 4.0954	D_m: 0.1496	D_p: 0.1405	Time: 12.5s
Epoch 066/100	G_loss: 3.9425	D_m: 0.1892	D_p: 0.1937	Time: 12.3s
Epoch 067/100	G_loss: 4.7198	D_m: 0.1019	D_p: 0.1160	Time: 12.5s
Epoch 068/100	G_loss: 3.7887	D_m: 0.1126	D_p: 0.1520	Time: 12.3s
Epoch 069/100	G_loss: 5.0482	D_m: 0.1763	D_p: 0.2026	Time: 12.4s
Epoch 070/100	G_loss: 4.7167	D_m: 0.1890	D_p: 0.2428	Time: 12.4s
Epoch 071/100	G_loss: 4.0714	D_m: 0.2256	D_p: 0.1817	Time: 12.4s
Epoch 072/100	G_loss: 5.0807	D_m: 0.1253	D_p: 0.2251	Time: 12.5s
Epoch 073/100	G_loss: 4.5828	D_m: 0.1556	D_p: 0.1943	Time: 12.3s
Epoch 074/100	G_loss: 3.5430	D_m: 0.1817	D_p: 0.1945	Time: 12.5s
Epoch 075/100	G_loss: 3.9404	D_m: 0.1403	D_p: 0.2448	Time: 12.4s
Epoch 076/100	G_loss: 4.5135	D_m: 0.0986	D_p: 0.1122	Time: 12.4s
Epoch 077/100	G_loss: 4.0981	D_m: 0.2319	D_p: 0.2348	Time: 12.5s
Epoch 078/100	G_loss: 3.9483	D_m: 0.1799	D_p: 0.1661	Time: 12.4s
Epoch 079/100	G_loss: 3.5998	D_m: 0.1586	D_p: 0.1770	Time: 12.7s
Epoch 080/100	G_loss: 3.8560	D_m: 0.1106	D_p: 0.2148	Time: 12.6s
Epoch 081/100	G_loss: 4.9922	D_m: 0.1989	D_p: 0.1328	Time: 12.5s
Epoch 082/100	G_loss: 5.3150	D_m: 0.0893	D_p: 0.1220	Time: 12.4s
Epoch 083/100	G_loss: 3.7756	D_m: 0.1408	D_p: 0.1876	Time: 12.3s
Epoch 084/100	G_loss: 3.6958	D_m: 0.2346	D_p: 0.1230	Time: 12.4s
Epoch 085/100	G_loss: 3.7964	D_m: 0.1060	D_p: 0.1987	Time: 12.2s
Epoch 086/100	G_loss: 4.0480	D_m: 0.1987	D_p: 0.1675	Time: 12.4s
Epoch 087/100	G_loss: 3.8413	D_m: 0.1246	D_p: 0.2099	Time: 12.3s
Epoch 088/100	G_loss: 3.6807	D_m: 0.1705	D_p: 0.1735	Time: 12.3s
Epoch 089/100	G_loss: 3.9471	D_m: 0.1321	D_p: 0.1680	Time: 12.5s
Epoch 090/100	G_loss: 4.5377	D_m: 0.2026	D_p: 0.2428	Time: 12.2s
Epoch 091/100	G_loss: 3.3404	D_m: 0.1693	D_p: 0.1595	Time: 12.5s
Epoch 092/100	G_loss: 4.3040	D_m: 0.1362	D_p: 0.1044	Time: 12.3s
Epoch 093/100	G_loss: 3.7080	D_m: 0.1664	D_p: 0.1778	Time: 12.4s
Epoch 094/100	G_loss: 3.4035	D_m: 0.3357	D_p: 0.1689	Time: 12.4s
Epoch 095/100	G_loss: 3.4141	D_m: 0.2024	D_p: 0.1470	Time: 12.3s
Epoch 096/100	G_loss: 4.1067	D_m: 0.2037	D_p: 0.1526	Time: 12.5s
Epoch 097/100	G_loss: 4.1465	D_m: 0.1756	D_p: 0.2019	Time: 12.3s
Epoch 098/100	G_loss: 3.7941	D_m: 0.1682	D_p: 0.1730	Time: 12.5s
Epoch 099/100	G_loss: 3.0862	D_m: 0.1860	D_p: 0.2002	Time: 12.4s
Epoch 100/100	G_loss: 3.6620	D_m: 0.1385	D_p: 0.2862	Time: 12.5s

This snippet creates a “submission_images” folder inside current output directory, then gathers and sorts all input photo paths. It loops over the first 10,000 images, loads and batches each one, runs it through the photo→Monet generator in inference mode, rescales the resulting tensor back to 0-255 uint8 pixel values, and saves each output as a zero-padded JPEG in the submission directory. Finally, it prints how many images were written.

```
In [21]: SUB_DIR = os.path.join(OUTPUT_DIR, "submission_images")
os.makedirs(SUB_DIR, exist_ok=True)
photo_paths = sorted(glob.glob(f"{photo_jpg_dir}/*.jpg"))

for i, path in enumerate(photo_paths[:10000]):
    img = load_image(path)[tf.newaxis,...]
    monet_img = G_photo2monet(img, training=False)[0]
    monet_img = ((monet_img + 1) * 127.5).numpy().astype(np.uint8)
    Image.fromarray(monet_img).save(f"{SUB_DIR}/{i:05d}.jpg")
```



```
print(f"Saved {min(len(photo_paths),10000)} images to {SUB_DIR}")
```

[illegible]

[illegible]

[illegible]

```

W0000 00:00:1745467861.724850    9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745467861.725216    9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745467861.725622    9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745467861.726168    9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745467861.727454    9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745467861.728254    9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745467861.729263    9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745467861.729898    9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745467861.730956    9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745467861.732348    9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced
W0000 00:00:1745467861.734245    9150 gpu_timer.cc:114] Skipping the delay k
ernel, measurement accuracy will be reduced

```

Saved 7038 images to cyclegan_output/submission_images

This code defines the source directory for the generated images and a target ZIP file path, then opens a new ZIP in write mode with DEFLATED compression. It walks through every file under submission_images, adds each JPEG to the archive using a relative path (so the ZIP is flat), and finally prints the path to the completed ZIP.

```

In [22]: src_folder = os.path.join(os.getcwd(), "cyclegan_output/submission_images")
zip_path = os.path.join(os.getcwd(), "generated/images.zip")

with zipfile.ZipFile(zip_path, "w", zipfile.ZIP_DEFLATED) as z:
    for root, _, files in os.walk(src_folder):
        for fname in files:
            full_path = os.path.join(root, fname)
            arcname = os.path.relpath(full_path, src_folder)
            z.write(full_path, arcname=arcname)

print("Done-created", zip_path)

```

Done-created /workspace/generated/images.zip

This code creates a 3×3 grid of 15×15-inch subplots at 100 DPI, then flattens the axes array so it can loop through the first nine Images to Monte images. For each subplot, it loads and rescales the image tensor from [-1, 1] back to [0, 1] for display, hides the axis ticks, and titles it with its image number. Finally, it applies a tight layout to remove extra padding and renders the figure.

```

In [23]: submission_jpgs = [os.path.join(src_folder, jpg) for jpg in os.listdir(src_f
fig, axes = plt.subplots(

```



```

nrows=3, ncols=3,
figsize=(15, 15),
dpi=100
)

axes = axes.flatten()
for ax, path in zip(axes, submission_jpgs[:9]):
    img = load_image(path)
    ax.imshow(((img + 1.0) / 2.0).numpy())
    ax.axis("off")
    ax.set_title(f"Image {submission_jpgs.index(path) + 1}")

plt.tight_layout()
plt.show()

```

Image 1



Image 2



Image 3

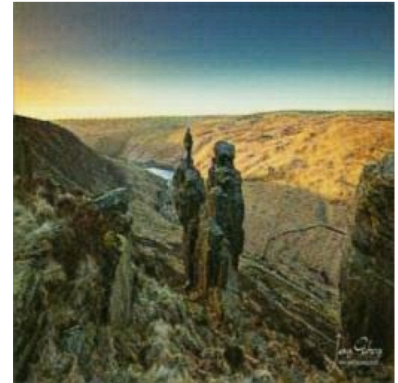


Image 4



Image 5

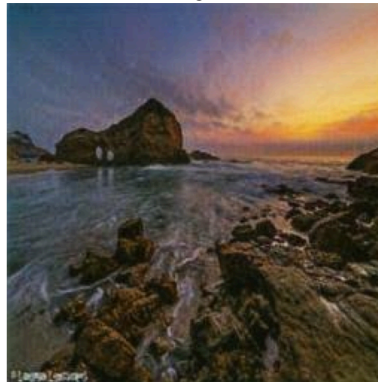


Image 6



Image 7

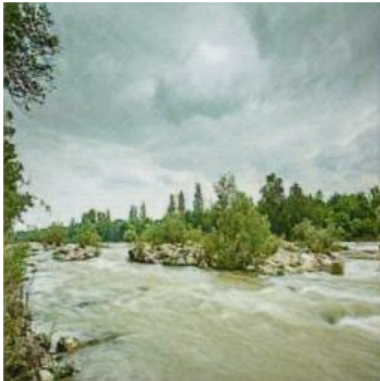


Image 8

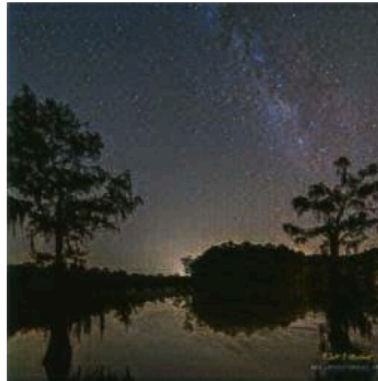


Image 9



Conclusion

In this project, I built and trained a CycleGAN to translate between two image domains—photos and Monet-style paintings—by defining modular data pipelines, generator and discriminator architectures, and a combined adversarial plus cycle-consistency loss. I set up efficient input handling with `tf.data` and AUTOTUNE, applied residual blocks in your generator for stable feature learning, and used a PatchGAN discriminator to focus on local texture realism. During training, I monitored both adversarial and cycle losses, periodically visualized sample translations, and ultimately generated a set of stylized images for submission. This end-to-end workflow—from preprocessing and model design to inference and packaging—demonstrates how CycleGANs can learn complex style mappings without paired data, and provides a solid foundation for further experimentation with loss weights, network depth, or alternative normalization strategies to enhance output quality.

References

1. [Amy Jang CycleGAN Tutorial](#)
2. [Geeksforgeeks GAN Tutorial](#)
3. [Tensorflow Docs](#)
4. [Keras API Docs](#)

This notebook was converted with convert.ploomber.io